Genes, Brain and Behavior
Official publication of the International Behavioural and Neural Genetics Society

Genes, Brain and Behavior (2015) **14:** 177–188 doi: 10.1111/gbb.12197

Borderline personality disorder and childhood maltreatment: a genome-wide methylation analysis

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Early life adversity plays a critical role in the emergence of borderline personality disorder (BPD) and this could occur through epigenetic programming. In this perspective, we aimed to determine whether childhood maltreatment could durably modify epigenetic processes by the means of a whole-genome methylation scan of BPD subjects. Using the Illumina Infinium® HumanMethylation450 BeadChip, global methylation status of DNA extracted from peripheral blood leucocytes was correlated to the severity of childhood maltreatment in 96 BPD subjects suffering from a high level of child adversity and 93 subjects suffering from major depressive disorder (MDD) and reporting a low rate of child maltreatment. Several CpGs within or near the following genes *(IL17RA, miR124-3, KCNQ2, EFNB1, OCA2, MFAP2, RPH3AL, WDR60, CST9L, EP400, A2ML1, NT5DC2, FAM163A and SPSB2)* were found to be differently methylated, either in BPD compared with MDD or in relation to the severity of childhood maltreatment. A highly relevant biological result was observed for cg04927004 close to *miR124-3* that was significantly associated with BPD and severity of childhood maltreatment. *miR124-3* codes for a microRNA (miRNA) targeting several genes previously found to be associated with BPD such as *NR3C1*. Our results highlight the potentially important role played by miRNAs in the etiology of neuropsychiatric disorders such as BPD and the usefulness of using methylome-wide association studies to uncover such candidate genes. Moreover, they offer new understanding of the impact of maltreatments on biological processes leading to diseases and may ultimately result in the identification of relevant biomarkers.

Keywords: Epigenetic, genetics, life events, miRNA, personality disorder

Received 1 October 2014, revised 01 December and 17 December 2014, accepted for publication 17 December 2014

Borderline personality disorder (BPD) is a severe psychiatric disorder affecting 2–4% of the general population (Lieb *et al.* 2004). This personality disorder is mainly characterized by marked impulsivity, poor emotion regulation and several behavioral dysfunctions, including suicidal acts. These dysfunctions lead to major impairments of interpersonal functioning and to poor socio-economic and familial outcomes.

One of the main etiological factors for the development of BPD is the existence of a history of childhood maltreatment (Paris 1994; Zanarini 2000). In this perspective, strong evidence points towards the main role played by early life adverse events, including childhood maltreatment, in the abnormal development of brain regions involved in the regulation of impulsivity and emotion (Driessen *et al.* 2000; Mccrory *et al.* 2013).

Child maltreatment may impact brain development by modifying epigenetic processes. The most robust findings in this field were obtained in the promoter regions of the glucocorticoid receptor (GR) gene (*NR3C1*). These studies found that when animals were exposed to early separation from their mother, they displayed a higher level of methylation of the promoter region of the *NR3C1* gene than animals normally reared (Mcgowan *et al.* 2011; Weaver *et al.* 2004). Several studies in humans investigated how early life adverse events impact on the methylation status of *NR3C1* promoter region (Mcgowan *et al.* 2009; Oberlander *et al.* 2008; Perroud *et al.* 2011). These studies consistently found an increased methylation status of the *NR3C1* promoter region in subjects, including BPD subjects, reporting early life adverse events (Martin-Blanco *et al.* 2014; Perroud *et al.* 2011; Steiger *et al.* 2013).

Although *NR3C1* has been the most investigated gene, hypo- or hyper-methylation of other genes are thought to play a role in mediating the impact of early life stress on the development of adulthood psychopathologies (Murgatroyd & Spengler 2011; Perroud *et al.* 2013; Thaler *et al.* 2014). In BPD subjects, childhood maltreatment and its severity correlated with higher methylation levels of two promoters of the *BDNF* gene (Perroud *et al.* 2013). Other studies found that BPD was associated with a different pattern of methylation of several genes including *HTR2A*, *MAOA*, *BDNF*, *DRD2* and *COMT* (Dammann *et al.* 2011; Groleau *et al.* 2014; Steiger *et al.* 2013; Thaler *et al.* 2014).

More recently, genome-wide methylation scans found broad reprogramming of DNA methylation in relation to

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early life adverse events and/or psychopathologies (Labonte *et al.* 2013). A study showed that subjects suffering from post-traumatic stress disorder (PTSD) and exposed to childhood maltreatment showed significantly higher changes in DNA methylation compared with those who were not exposed to these events (Mehta *et al.* 2013). Teschler *et al.* (2013) in a genome-wide methylation analysis found several specific CpG sites that exhibited increased methylation in BPD patients compared with controls.

We here performed a genome-wide methylation scan in patients suffering from BPD and a control sample of depressed subjects with low rate of child maltreatment in order to identify new genes that may be involved in the cascade of events leading from child maltreatment to the development of BPD.

Materials and methods

Subjects

A total of 96 patients suffering from BPD were recruited in a specialized center for the treatment and care of this disorder. All patients were assessed for Axis I psychiatric disorders using the diagnostic interview for genetic studies (DIGS) (Preisig *et al.* 1999) and completed several self-report questionnaires assessing dimensions closely related to BPD, such as impulsiveness and emotion regulation. Subjects were assessed for BPD using the structured clinical interview for DSM-IV personality disorders (SCID-II) (First *et al.* 1994), and those with a score below 5/9 for the SCID-II BPD part were excluded from the study. History of childhood maltreatment was assessed using the Childhood Trauma Questionnaire (CTQ) which assesses five types of trauma (sexual abuse, emotional abuse and neglect, physical abuse and neglect). Each trauma has a score of severity with four options ranging from none, low, moderate to severe according to Bernstein and Fink (1998). Moreover, the self-reported history of maltreatment was discussed and validated with the patient during a clinical interview, as previously described (Perroud *et al.* 2011).

As BPD subjects were expected to show a high rate of abuse and neglect (thus methylation in subjects with no maltreatment would be difficult to ascertain because of the ceiling effect), 93 patients suffering from major depressive disorders (MDDs) were recruited from consecutive admissions to the psychiatric university hospitals of Montpellier (France) and Geneva (Switzerland). These subjects were thus selected for their low rate of child abuse and neglect among a cohort of more than 1500 subjects with available CTQ as previously described (Perroud *et al.* 2011). Moreover, in order to increase the likelihood of comparing them to BPD subjects, these subjects were chosen as having psychological dimensions clearly distinguishing them from BPD subjects (low impulsiveness as measured by the Barrat Impulsiveness Scale (BIS-10) (Bayle *et al.* 2000), low anger traits as measured by the State-Trait Anger Expression (STAXI) (Spielberger 1998), and low harm avoidance as measured by the Temperament and Character Inventory [TCI; (Cloninger *et al.* 1993)]. Most of them were suicide attempters ($N = 90$) and are thus ideally suited to control for this comorbidity frequently found in BPD in our analyses. Of note 70 (72.9%) of our BPD subjects had also a history of suicide attempt. Moreover, 72 of them (75%) were also suffering from comorbid MDD giving additional rational for the use of MDD subjects as control group in the context of our study [see (Perroud *et al.* 2011) for detailed description of the samples used in this study].

The study was approved by the ethical committees in each center and all subjects gave their informed written consent.

DNA extraction and methylation scans

For the BPD subject, blood samples were systematically collected 1 week before the start of the treatment. For MDD subjects, blood samples were collected just after the interviews. DNA was extracted from blood white cells using the Nucleon extraction kit (Amersham Life Science, Dubendorf, Switzerland).

Genome-wide methylation was assessed using the Illumina Infinium HumanMethylation450 measuring 485 577 methylation sites (CpGs).

Pyrosequencing analysis of DNA methylation at cg04927004 (near miR124-3)

As being the most relevant and consistent finding of our study, we performed a pyrosequencing analysis of DNA methylation at cg04927004 (near miR124-3). Assay design to measure methylation level in cg04927004 with bisulfite-pyrosequencing was performed using MethPrimer software (Li & Dahiya 2002). Cg04927004 is located within a 4613 bp CpG island composed of 424 CpG also containing one of the three loci of miR-124, called here miR-124-3. The starting sequence of cg04927004 was obtained from the Illumina datasheet providing information about the position and the sequence context of that CpG site. The exact position of cg04927004 is on chromosome 20 at coordinates 61′809′348 when using the hg19 genome. The assays designed to measure methylation levels at cg04927004 consist of the primers and sequence information provided in the table. The assays was performed in duplicates on 47 individuals belonging to the MDD and BPD subjects that were analysed in the 450 K arrays (29 BPD and 18 MDD). For these individuals, bisulfite conversion of unmethylated cytosines to uracil was performed on 2 μg of DNA using the EpiTect Bisulfite Kit (Qiagen, Hilden, Düsseldorf, Germany). Note that the bisulfite treatment was thus independent experimentally as compared with the 450 K experiment. Bisulfite treated samples were then resuspended in 20 μl EB Buffer and were stored at −20°C. PCR reactions were performed in 25 μl final volume containing 50 ng bisulfite-treated DNA, 1× PCR buffer, 1.6 mM MgCl2, 200 μM dNTP, 0.2 μM forward primer TYGGGATTGGTAATTAYGTT (Y consist of an equivalent number of primer molecules carrying either C or T), 0.2 μM reverse primer 5′ Biotine-AACTACCTTCCCCTACCRTC (R consist of an equivalent number of molecule carrying either A or G), 2 units of Hot Start Taq DNA polymerase (HotStart Taq DNA Polymerase Kit, Qiagen). PCR condition was as follows: 95°C for 15 min followed by 50 cycles of 95°C for 30 seconds, 51°C for 30 seconds, 72°C for 10 seconds resulting in a 254 bp PCR products. That PCR product was immobilized onto streptavidin-coated sepharose beads (Fisher Scientific, Reinach, Switzerland, ref 17-5113-01) using a Vacuum preparation tool (Biotage, Uppsala, Sweden), washed in ethanol 70%, and denatured in 0.2 M NaOH. Single stranded DNA molecules were then washed in 10 mM Tris-acetate pH 7.6 before being dispensed into 20 mM Tris, 2 mM $MgAc_2$ containing 15 pmol of the forward primer used as a sequencing primer in 40 μl final volume by gently agitating the aspirator. The resulting single stranded DNA molecules were then sequenced in a PSQ 96 MA (Qiagen) using appropriate enzymes, substrates and nucleotides (PyroMark Gold Q96 SQA Reagents, Qiagen). Percentages of methylation were obtained according to the ratio of pics obtained after cytosine and thymine sequential dispensation at the CpG site position that is automatically calculated by the software (see Table S1, Supporting Information).

Statistics

Raw microarray data from Illumina arrays were exported using the software GenomeStudio version 2009.1 and further processed under R environment.

Data preprocessing and quality control

Preprocessing consisted in normalizing intensity signals measured by the instrument, and updating subsequent methylation ratios following the procedure implemented by the R-package 'methylumi' Davis *et al.* (2013). Corrected methylation ratios were further centered at each CpGs site (i.e. we subtract the mean value observed across all samples at this site), and ratios associated with a P value > 0.01 were replaced by zeroes. If more than 10 samples met the P value criteria at one CpG site, the site was removed from the dataset. This minimal call rate above 95% removed 4658 sites out of the initial 485 577

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sites. A total of 81 CpG sites located on sexual chromosomes Y were also removed, so that only chromosomes 1–22 and chromosome X remained. One individual was removed for low call rate below 95%.

BPD and MDD subjects were analysed on two separate chips. In order to account for the batch effects, three BPD subjects from the first chip were added to the second one as duplicates. As expected, the average difference of methylation between the two batches was close to 0%, with a standard deviation of ~4.7%. Following this result, CpGs where methylation ratios of one duplicate varied by more than 10% between batches were removed from the analyses ($N = 55\,240$ CpGs).

Altogether, preprocessing and quality control steps reduced the size of the dataset from 485 577 initially measured CpGs to 425 533 CpGs.

Correcting for blood cell types

Human blood cells have a different pattern of methylation depending on their type (Reinius *et al.* 2012). It therefore follows that total blood composition with different proportion of cell types can represent a major source of variability for methylation analyses (Jaffe & Irizarry 2014). Bioinformatics tools have been designed to estimate the proportion of cell types in a whole blood sample by comparing its methylation profile to a database of cell sorted profiles (Houseman *et al.* 2012; Jaffe & Irizarry 2014). These tools have become essential to assert that variations in blood composition between samples are not confounding factors for the study. To check for blood cell type composition, the data were processed with R package minfi v1.10.1 (Aryee *et al.* 2014). Raw signal intensities for red and green channels were directly loaded from idat files, merging all the samples from both batches. Cell proportion estimation was carried out by a call to the method '*estimate Cell Counts*' (Jaffe & Irizarry, 2014). A *t*-test was further applied to check for the existence of a significant difference of composition between BPD and MDD samples. Estimation obtained on the three replicated BPD samples was used as a quality control for the estimation. In addition, as gender might play a role in this comparison, we also tested for methylation differences on female subjects only, and between men and women in MDD subjects. Finally, the database of cell-sorted methylation profiles was queried at the CpG sites of interest to assert that they were not individually more represented into one cell type than into another. The R package *FlowSorted.Blood.450k* v1.2.0 was used as database source. Raw data generated for this study were uploaded to GEO repository (Gene Expression Omnibus) on NCBI.

Multivariate and univariate analyses

Multivariate analysis. The multivariate analysis consisted in training a linear Support Vector Machine (SVM) model with L1-regularization in order to predict the binary outcome (BPD vs. MDD) based on methylation values. The training process was carried out on a subset of the normalized dataset built by randomly choosing 115 female samples, and the three duplicates (so 121 training instances in total). The 71 remaining samples were not used for model fitting, but as an independent validation set for an unbiased estimate of model performances. Duplicated samples were forced to be included in the training set in order to capture their similitude and secondarily correct for batch effect. Accordingly, we modified the L1-SVM algorithm in order to penalize differences in prediction on replicated samples. In addition, we added more stringency (five times more stringent) to the prediction error tolerance for those replicated samples compared with other samples.

Parameter LAMBDA of L1-SVM algorithm was used to control sparseness of the generated linear models. We manually tuned the parameter to a value of 6.0 observing that it generates a linear model with only 30 CpGs with non-zero linear coefficients. The other 425 503 variables had a linear coefficient of 0, and thus did not play any role in the decision function. This considerable dimensionality reduction to a subset of 30 CpGs was a choice to avoid over-fitting the training set.

For the analyses of CTQ, the same normalization procedure than for the case–control study was used, as well as the same dataset (i.e. 121 female for training and the 71 remaining for testing). We then fit an ordinal regression model, with L1 regularization to capture the information contained in the data. An ordinal regression model is a linear prediction model used to compare the methylation values of two patients and predict the one with highest CTQ score. Such a model was fitted for each of the five types of trauma treated as ordinal variables with scores ranging from 1 to 4 (none, low, moderate and severe). In addition, we included a 6th ordinal category, ranging from 0 to 5, obtained by counting the number of CTQ categories with an existing severity level (i.e. either low, moderate or severe) as previously described (Perroud *et al.* 2011, 2013). This last category can be observed as a global score of childhood maltreatment obtained by combining the five standard categories (= 'global CTQ'). Prediction performance was measured by counting the number of correct predictions made by the models when comparing all pairwise combinations of testing samples. Significance of the prediction performance was measured by comparing those counts to the ones obtained by 1000 random predictors that uniformly order the instances.

Univariate analysis. A *t*-test was used to assert the significance of methylation difference between MDD and BPD. To avoid the influence of the gender variable, only the 150 female subjects of the dataset were considered. In addition, to avoid false positive hits related to batch effects, we considered only CpG sites where the three duplicated samples differ by less than 10% between the two batches. Finally, the average methylation values of MDD patients was compared with the average methylation value of BPD patients, and the difference was required to be higher than 10%.

For the analysis of the CTQ, a Spearman correlation test was used to assert the significance of the correlation between methylation of a CpG site and CTQ scores. To get rid of gender effect, the analyses were restricted to the women. In addition, to avoid false positive hits related to batch effects, only CpGs where the three duplicated samples differed by less than 10% between the two batches were considered. Finally, the average methylation values of patients with the highest 'global CTQ' (= 5) was compared with the average methylation value of the patients with the lowest 'global CTQ' (= 0), and the difference was required to be higher than 10%.

For all the significant results, age was added as a confounding variable.

Functional analysis. The software Cytoscape 2.8.3 was used to retrieve known gene–gene interactions from the IntAct database. The interaction network was further simplified to keep only nodes interacting with at least two other genes.

Enrichment analysis was performed using the web service DAVID.

Results

Table 1 displays the clinical and demographic characteristics of the sample.

Blood cell types

The result of blood cell type estimation is summarized in Fig. 1. On average, granulocytes represented 58% of the whole blood content, while the five other types represented less than 15%. Those results are in agreement with the values reported by (Jaffe & Irizarry 2014), which aggregate the data of 1098 samples from five independent studies on whole blood. Furthermore, if we compare estimations on our three replicated samples, there is on average a difference of 5% on granulocytes proportion estimate.

Looking at the difference between BPD and MDD subjects, BPD had significantly more CD8T cells ($t = 6.2$, $P = 2.7 \times 10^{-9}$), significantly less monocyte cells ($t = -3$, $P = 0.002$) and less CD4T cells ($t = -2.3$, $P = 0.02$). However,

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Table 1: Clinical and demographic characteristics of the sample

	BPD ($n=96$)		MDD ($n=93$)		
	Mean	SD	Mean	SD	P (t-test)
Age	32.17	0.99	41.39	1.37	<0.001
	N	%	N	%	P (χ^2)
Gender (Female)	88	91.67	59	63.44	<0.001
Past/current alcohol use disorder	52	54.17	17	18.28	<0.001
Past/current substance use disorder	43	44.79	3	3.23	<0.001
Taking a psychopharmacological treatment	73	51.05	70	48.95	0.90
Sexual abuse	53	55.21	0	0	NA
Physical abuse	46	47.91	0	0	NA
Physical neglect	54	56.25	0	0	NA
Emotional abuse	81	84.37	0	0	NA
Emotional neglect	79	82.29	49	52.69	<0.001

For childhood maltreatment, percentage stands for presence of maltreatment.

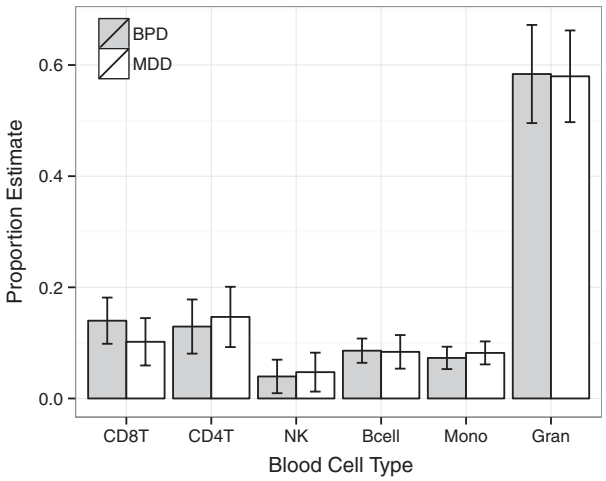

Figure 1: Comparison of cell type proportion between BPD and MDD samples. Bcell: B-cell; CD8T: CD8+ T-cell; CD4T: CD4+ T-cell; Error bars = standard deviation (SD); Gran: granulocyte; Mono: monocyte; NK: natural killer cell.

there was no significant difference between granulocytes, which was the main component of our samples. Considering women subjects only, the significances were less pronounced (CD8T: $t=4.0$, $P=7\times10^{-5}$, monocytes: $t=-2.8$, $P=0.005$; CD4T: $t=-1.8$, $P=0.06$). Furthermore, in the MDD population, we observed that women had significantly more CDT8 cells than men ($t=3.6$, $P=5\times10^{-4}$), which in part explained the difference observed between BPD and MDD subjects on the whole population.

Case–control study

Multivariate analyses

On the training set, the model showed an accuracy of 93.3%, and it correctly predicted all duplicated samples. On the testing set, accuracy decreases to 84.5%, still significantly better than a random classification with a P value lower than 1×10^{-6}.

Variables with non-zero coefficients in the model are listed in Table 2. The CpG sites with the strongest coefficient, and therefore the greatest weight in the decision function were: cg04907664 on Xq13.1 within the gene coding for a type I membrane protein and a ligand of Eph-related receptor tyrosine kinases (*EFNB1*) ($\beta=1.7$), cg16637873 on 12p13.31 localized close to the gene coding for a suppressor of cytokine signaling (SOCS) family member (*SPSB2*) ($\beta=-1.57$) and cg11897887 on 20p11.21 localized 1500 bp downstream of the gene coding for a protein similar to mouse cystatin 9 (*CST9L*) ($\beta=-0.91$). Moreover, several CpGs were found on the same chromosomal regions, including 13q34, 2p25.3, 5p15.33 and 9q34.2, which suggest that the model has captured broader signal coming from these specific regions.

Univariate analyses

Table 3 lists the 20 top CpGs according to the t-test. The three most significant results were: cg04927004 on 20q13.33 close to the gene coding for the microRNA 124 (*miR124-3*) with lower methylation status in BPD than in MDD [0.22 (SD = 0.04) vs. 0.35 (SD = 0.04), $P=3.35\times10^{-39}$], cg07430978 on 7q36.3 1500 bp upstream from the gene coding for WD repeat domain 60 (*WDR60*) with lower methylation status in BPD than in MDD [0.59 (SD = 0.03) vs. 0.69 (SD = 0.03), $P=5.15\times10^{-37}$] and cg06656994 on 1q25.2 in the 5′ UTR region of the family with sequence similarity 163, member A (*FAM163A*) also with lower methylation status in BPD than in controls [0.13 (SD = 0.03) vs. 0.25 (SD = 0.04), $P=5.49\times10^{-35}$]. A higher number of significant CpGs were localized on chromosome X with more methylated sites in BPD subjects and on chromosome 6 (6p22.1, 6p21.33 and 6p21.32) with higher methylated CpGs in MDD subjects than in BPD patients (Table 3).

Effect of child maltreatment

Multivariate analyses

Prediction performances were relatively good for three CTQ categories (emotional abuse and neglect and sexual abuse) with the highest performance obtained for emotional abuse. The 'global' model (6th ordinal category) gives the highest

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Table 2: CpG sites with non-zero linear coefficients

CpG	Position (bp)	Chromosome	Gene	β	*t*-test	μBPD	μMDD
cg04907664	X:68049805	Xq13.1	EFNB1[Body]	1.701	1.81×10^{-23}	0.61	0.4
cg16637873	12:6982822	12p13.31	SPSB2[TSS1500]	−1.574	0.0022	0.05	0.2
cg11897887	20:23550632	20p11.21	CST9L[TSS1500]	−0.908	0.0023	0.55	0.47
cg04998327	11:103453960	11q22.3	Intergenic	−0.831	0.1033	0.53	0.61
cg08616760	5:178157801	5q35.3	ZNF354A[TSS200]	0.69	8.75×10^{-32}	0.31	0.19
cg13935315	19:12547760	19p13.2	ZNF443[Body]	0.516	0.0142	0.68	0.77
cg27625131	13:113105794	13q34	SPACA7; TUBGCP3	−0.482	0.0537	0.52	0.49
cg10724632	2:3675155	2p25.3	COLEC11[Body]	0.419	0.0458	0.33	0.42
cg14417873	9:136838534	9q34.2	VAV2[Body]	0.395	0.0008	0.6	0.72
cg27292220	6:30980847	6p21.33	MUC21/…	−0.376	0.0007	0.73	0.64
cg06520095	16:66458043	16q21	BEAN1/CDH5	0.298	0.0044	0.61	0.52
cg01957222	2:1426786	2p25.3	TPO[Body]	−0.29	0.0094	0.73	0.66
cg04927004	20:61809348	20q13.33	MIR124-3[TSS1500]	0.282	3.35×10^{-39}	0.22	0.35
cg05619116	3:27754569	3p24.1	EOMES	0.279	8.60E−14	0.08	0.26
cg01741056	16:25075152	16p12.1	LCMT1/ARHGAP17	0.273	0.0024	0.7	0.78
cg21874902	5:33727007	5p13.3	ADAMTS12[Body]	−0.223	0.0008	0.8	0.71
cg08506672	5:3959743	5p15.33	IRX1(350 kb)	−0.2	0.0114	0.41	0.34
cg13573375	19:4033718	19p13.3	PIAS4[Body]	0.196	0.0545	0.47	0.52
cg05179499	16:83982669	16q23.3	OSGIN1[TSS200]	0.192	0.0079	0.64	0.71
cg18275316	5:3607063	5p15.33	IRX1	0.188	7.61×10^{-31}	0.14	0.28
cg13972557	5:116075820	5q23.1	RPS17P2	−0.184	0.1932	0.53	0.44
cg14797147	5:124309096	5q23.2		−0.181	0.3479	0.56	0.6
cg20089799	12:3384898	12p13.32	TSPAN9[Body]	0.178	0.0258	0.44	0.37
cg08880261	21:43547872	21q22.3	UMODL1[Body]	0.088	0.0124	0.5	0.57
cg15567368	7:563891	7p22.3	PDGFA	0.084	0.0443	0.43	0.37
cg13564529	9:136822302	9q34.2	VAV2[Body]	−0.078	0.0457	0.58	0.49
cg13653328	5:148520669	5q32	ABLIM3 × [TSS1500]	−0.065	0.2457	0.58	0.51
cg07878625	7:148977647	7q36.1	ZNF783[Body]	−0.053	0.0044	0.63	0.74
cg16595484	3:122512170	3q21.1	HSPBAP1[Body]	−0.035	1.4×10^{-11}	0.13	0.3
cg26512469	13:113688268	13q34	MCF2L[Body]	−0.005	0.0044	0.83	0.74

The sites are ordered by increasing linear coefficient value (β) of the multivariate model. The *t*-test column indicates the result of the univariate *t*-test comparing MDD and BPD methylation values. μBDL (resp. μMDD) is the mean methylation ratio for BPD (resp. MDD) subjects at the given site. TSSxxx = CpG is located within xxx bp of the transcription start site.

performances and was thus chosen as the main outcome in the following analyses. Table 4 shows the CpGs associated with the 'global CTQ'. 10 CpGs displayed a coefficient $\beta >= 1$, indicating a strong influence in the prediction model. These CpG sites were located within or near the following genes: human homologue of the mouse p (pink-eyed dilution) gene (*OCA2*); microfibrillar-associated protein 2 (*MFAP2*); cystatin 9-like (*CST9L*); E1A binding protein p400 (*EP400*); potassium voltage-gated channel, KQT-like subfamily, member 2 (*KCNQ2*); alpha-2-macroglobulin-like 1 (*A2ML1*); 5′-nucleotidase domain containing 2 (*NT5DC2*) and rabphilin 3A-like (*RPH3AL*).

Univariate analysis

Table 5 lists the 20 top CpG according to the Speerman correlation test. The most significant signals came from cg26298038 on 22q11 ($P = 3.62 \times 10^{-21}$) in the 1st Exon (5′ UTR) of the gene coding for the interleukin 17 receptor A (*IL17RA*), cg26196213 in an intergenic region on 6p22.1 ($P = 2.05 \times 10^{-20}$), and cg04927004 on 20q13.33 1500 bp upstream *miR124-3*. Of note the most significant result according to the univariate analyses appearing in the top results of the multivariate analyses was *miR-137* (Table 4).

Variable shared by models

Some CpGs were shared by the different models and were thus associated with both disease status and severity of childhood maltreatment, but only one was found in three out of the four analyses above: *miR124-3* (Fig. 2). In other to ensure that this CpG was truly associated with the outcomes, we checked if it might have been confounded by the blood cell types. Table 6 gives the result of querying the sorted blood cell database for cg04927004 (*miR124-3*). The mean methylation of cg04927004 was 12% higher in CD8T cell than in granulocytes. This bias, in regard to the difference of composition found between BPD and MDD subjects on CD8T cells, might explain a higher methylation in BPD subjects than in MDD subjects on this site. However, we observed exactly the opposite in our data: BPD subjects had a lower methylation than MDD subjects. So blood composition should definitely not represent a confounding factor for this site.

The known targets of *miR124-3* are listed in Table 7.

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Table 3: 20 top CpGs according to the *t*-test comparing BPD to MDD subjects

CpG	Position (bp)	Chromosome	Gene	*t*-test	μBPD	σBPD	μMDD	σMDD
cg04927004	20:61809348	20q13.33	MIR124-3[TSS1500]	3.35×10^{-39}	0.22	0.04	0.35	0.04
cg07430978	7:158647784	7q36.3	WDR60[TSS1500]	5.15×10^{-37}	0.59	0.03	0.69	0.03
cg06656994	1:179713176	1q25.2	FAM163A[5′UTR]	5.49×10^{-35}	0.13	0.03	0.25	0.04
cg14282508	14:103546492	14q32.32		2.04×10^{-33}	0.65	0.02	0.75	0.04
cg18101118	3:45017337	3p21.31	EXOSC7[TSS1500]; ZDHHC3[5′UTR]	6.88×10^{-33}	0.33	0.05	0.21	0.04
cg03406394	15:60296138	15q22.2	FOXB1[TSS1500]	3.56×10^{-32}	0.09	0.02	0.2	0.04
cg08616760	5:178157801	5q35.3	ZNF354A[TSS200]	8.75×10^{-32}	0.31	0.05	0.19	0.04
cg18275316	5:3607063	5p15.33		7.61×10^{-31}	0.14	0.05	0.28	0.06
cg00350607	6:159057720	6q25.3	DYNLT1[3′UTR]	3.66×10^{-30}	0.7	0.03	0.58	0.05
cg01245787	4:71859630	4q13.3	DCK[1st Exon]	3.67×10^{-30}	0.08	0.02	0.18	0.04
cg22502459	8:141108998	8q24.3	TRAPPC9[Body]	4.52×10^{-29}	0.67	0.05	0.56	0.04
cg05529448	6:12009163	6p24.1		1.4×10^{-28}	0.24	0.04	0.14	0.04
cg22174410	3:42641766	3p22.1	NKTR[TSS1500]	1.52×10^{-28}	0.07	0.03	0.19	0.05
cg14977069	20:62367698	20q13.33	LIME1[TSS1500]	1.86×10^{-28}	0.69	0.05	0.57	0.05
cg23271606	7:92464891	7q21.2	CDK6[5′UTR]	7.65×10^{-28}	0.1	0.02	0.2	0.04
cg15825928	6:126964945	6q22.32		1.1×10^{-27}	0.71	0.05	0.61	0.04
cg01235203	3:187461998	3q27.3	BCL6[5′UTR]	4.46×10^{-27}	0.09	0.02	0.21	0.05
cg08133824	7:99698538	7q22.1	MCM7[Body, TSS200]; AP4M1[TSS1500]	5.87×10^{-27}	0.05	0.02	0.16	0.05
cg16327326	16:1662895	16p13.3	IFT140[TSS1500]	7.46×10^{-27}	0.09	0.02	0.2	0.05
cg00973947	3:143692255	3q24	C3 or f58[5′UTR, Body, 1st Exon]	1.16×10^{-26}	0.09	0.02	0.21	0.05

TSSxxx = CpG is located within xxx bp of the transcription start site.

Functional analysis

The gene *GRB2* (growth factor receptor-bound protein 2) and the 'Chemical Entities of Biological Interest' CHEBI: 17283 (1-phosphatidyl-1D-myo-inositol 3-phosphate) were found to interact with several of the genes found to be associated either with BPD or with severity of childhood maltreatment. The enrichment analysis highlighted five main clusters: SH3 domain (*RIMBP3*, *BIN1*, *VAV2* and*MCF2L*); apoptosis and programming of cell death (*PCSK9*, *ZNF443*, *VAV2* and *MCF2L*); EGF-like domain (*UMODL1*, *CNTNAP5* and *TPO*); GTPase regulator activity (*RPH3AL*, *ARHGAP17*, *VAV2* and *MCF2L*); and cation/ion binding (*ZNF354A*, *ADARB2*, *RPH3AL*, *NT5DC2*, *COLEC11*, *VAV2*, *ZZEF1*, *CDH5*, *PIAS4*, *UMODL1*, *ZNF443*, *PCSK9*, *TPO*, *ADAMTS12* and *KCNQ2*).

Pyrosequencing analysis of DNA methylation at cg04927004

The results of the pyrosequencing on selected individuals showed higher level of methylation at cg04927004 in MDD compared with BPD subjects (see Fig. S1). These results thus validate the ones found in the univariate analysis in the whole methylation scan (see above) showing lower methylation status in BPD than in MDD for cg04927004. Moreover the correlation of the results obtained by the Illumina chip and the pyrosequencing showed a positive correlation of 0.27 (see Fig. S2).

Discussion

In this study, we found that early life adverse events, namely severity of childhood maltreatment as well as BPD, were associated with several CpGs located within or near genes involved in several biological functions such as inflammatory processes (*IL17RA*), regulation of gene expression (*miR124-3*), neuronal excitability and development or maintenance of the nervous system (*KCNQ2* and *EFNB1*), structure of cell and regulation of cell functions such as molecule transport through cell membranes, calcium-ion-dependent exocytosis in both endocrine and exocrine cells, cycle progression, signal transduction, apoptosis and gene regulation (*OCA2*, *MFAP2*, *RPH3AL* and *WDR60*), and other biological functions (*CST9L*, *EP400*, *A2ML1*, *NT5DC2*, *FAM163A* and *SPSB2*). Several of these genes have previously been found to be associated with neuropsychiatric disorders, such as *KCNQ2*, which is involved in bipolar disorder (Judy *et al.* 2013), or *NT5DC2*, previously found to be associated with schizophrenia (Ripke *et al.* 2013). In addition, *NT5DC2* is a target of *miR137*, one of the genes found to be associated with childhood maltreatment in our sample (see Table 4). The most relevant CpG in our study was cg04927004 close to the gene coding for a microRNA (*miR124-3*), which was found to be significantly associated, in our sample, with both severity of childhood maltreatment and BPD, and may therefore be involved in the path leading from early life adverse events to the disorder in adulthood. MicroRNAs (miRNAs) are short non-coding RNAs (21–23 nucleotides) regulating protein synthesis through translational repression or mRNA degradation, and are thus key actors in regulating the environmental impact on genetic expression. As miR124, miRNAs are highly expressed in neural cells and because they are considered as the main regulator of neural plasticity and neurogenesis, they appear to be a strong candidate for neuropsychiatric disorders (Dwivedi 2014). Several studies have indeed found an altered expression of miRNAs in the brain and peripheral

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Table 4: List of CG sites selected for their relative contribution to 'global CTQ', and their respective Spearman correlation test (*P* value)

CpG	Position	Chromosome	Gene	β	Correlation *P* value
cg03449867	15:28200653	15q13.1	OCA2[Body]	−2.18	0.027
cg15600437	1:17309539	1p36.13	MFAP2[TSS1500]	−1.785	0.0281
cg11897887	20:23550632	20p11.21	CST9L[TSS1500]	−1.654	0.0407
cg06813297	12:132537251	12q24.33	EP400[Body]	−1.368	0.0196
cg00017157	1:167127353	1q24.1		1.177	0.0029
cg13379325	20:62052259	20q13.33	KCNQ2[Body]	1.164	0.0017
cg16435686	10:47964834	10q11.22		1.125	0.0002
cg24309769	12:8995965	12p13.31	A2ML1[Body]	1.09	0.0026
cg22710716	3:52569097	3p21.1	NT5DC2[TSS200, TSS1500]	−1.022	0.1175
cg08900396	17:149582	17p13.3	RPH3AL[Body]	−1.011	0.0815
cg03526459	1:146549940	1q21.1		0.969	0.0011
cg16069065	7:129412730	7q32.2		−0.896	0.0916
cg18828306	11:17555864	11p15.1	USH1C[Body]	−0.774	0.0531
cg27625131	13:113105794	13q34		0.737	0.0055
cg11036359	6:29759078	6p22.1	HCG4[Body]	−0.688	0.0212
cg02887598	2:127841945	2q14.3	BIN1[Body]	0.674	0.0822
cg04417708	17:4043867	17p13.2	ZZEF1[Body]	−0.632	0.0527
cg06352616	16:1493759	16p13.3	CCDC154[Body]	0.609	0.5585
cg14035771	1:98515272	1p21.3	MIR137[TSS1500]	0.588	3.29×10^{-11}
cg27586797	5:13664584	5p15.2		−0.556	0.652
cg01013522	2:124971544	2q14.3	CNTNAP5[Body]	0.459	0.1287
cg08506672	5:3959743	5p15.33		0.354	0.0337
cg21874902	5:33727007	5p13.3	ADAMTS12[Body]	0.354	0.0002
cg19285525	2:161209326	2q24.2	RBMS1[Body]	−0.334	0.0155
cg20859841	8:1400221	8p23.3		−0.327	0.1638
cg00045070	1:55504649	1p32.3	PCSK9[TSS1500]	0.314	0.9306
cg00944631	22:20464947	22q11.21		0.299	0.0228
cg05818501	10:1451648	10p15.3	ADARB2[Body]	0.272	0.0003
cg05619116	3:27754569	3p24.1		0.118	5.16×10^{-11}
cg20073472	6:31146845	6p21.33	PSORS1C3[TSS1500]	−0.117	6.54×10^{-6}
cg14112997	22:17591088	22q11.1	IL17RA[3′UTR]	0.06	0.0015

Sites are sorted according to the absolute value of their linear coefficient (β), so that it reflects their importance in the multivariate model. TSSxxx = CpG is located within xxx bp of the transcription start site.

blood of subjects suffering from psychiatric disorders (Smalheiser *et al.* 2014). Moreover, recent studies have suggested that epigenetics might be at play in the regulation of miRNAs (Chuang & Jones 2007). Conversely, miRNAs are able to modulate the expression of epigenetic regulators and thus indirectly modulate gene expression (Sato *et al.* 2011). As some miRNAs have been shown to be expressed at different developmental stages, they are of particular interest for the understanding of the impact of early life adverse events on brain development and in the link with adulthood psychopathologies such as BPD (Dwivedi 2014). It has recently been shown that early life stress alters miRNAs expression in mice, as well as their behavior and metabolic response. Indeed, in a recent study, Gapp *et al.* (2014) found that, in mice, traumatic stress endured early in life durably impacts gene expression of miRNAs resulting in behavioral and metabolic alterations in offspring. Moreover in a recent methylome-wide association study of schizophrenia, Aberg *et al.* (2014) found that several miRNAs were potentially involved in the etiology of this disorder. Concerning miR124, it is the more abundant miRNA in neurons and has been shown to closely interact with cAMP response element-binding (CREB) protein, a protein involved in the etiology of several psychiatric disorders (Saab & Mansuy 2014). This miRNA is of much importance for neuronal differentiation, neurogenesis and more interestingly targets two crucial genes in the regulation of the Hypothalamic–pituitary–adrenal (HPA) axis (*NR3C1 and NR3C2*), genes for which the methylation status was shown to be modified in relation to either BPD or adverse life events (Perroud *et al.* 2011, 2014; Yu *et al.* 2008). Besides *NR3C1* and *NR3C2*, miR124 also targets the gene *EFNB1*, which is also a gene affected by the top-ranked site in our case–control study. Interestingly and closely related to BPD, recent data in rats has found miR124 upregulation in the lateral amygdala following fear conditioning (Griggs *et al.* 2013). As amygdala hyperactivity in reaction to fearful faces has been consistently found in BPD subjects, it would be interesting to directly test whether miR124 is upregulated in BPD patients with amygdala hyperactivity (Krause-Utz *et al.* 2014). Our data support this hypothesis, given that a lower methylation level of a CpG upstream of *miR124* was found in BPD patients compared with MDD subjects. Cao *et al.* (2013) found downregulation of miR124 in the hippocampus of rats exposed to unpredictable chronic mild stress and separation. Our results thus support

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Table 5: 20 top CpsG according to the Spearman correlation test testing association with 'global CTQ'

CpG	Position	Chromosome	Gene	Correlation test *P* value
cg26298038	22:17565904	22q11.1	IL17RA[1st Exon, 5′UTR]	3.62×10^{-21}
cg26196213	6:29617956	6p22.1		2.05×10^{-20}
cg04927004	20:61809348	20q13.33	MIR124-3[TSS1500]	1.18×10^{-19}
cg13390975	5:34915890	5p13.2	BRIX1[1st Exon]; RAD1[TSS200]	3.48×10^{-19}
cg13610241	3:184032169	3q27.1	EIF4G1[TSS200, TSS1500]	5.77×10^{-18}
cg16317734	11:94822598	11q21	ENDOD1[TSS1500]	7.96×10^{-18}
cg09691861	4:77870047	4q21.1	SEPT11[TSS1500]	1.81×10^{-17}
cg21682382	1:32538052	1p35.1	TMEM39B[TSS1500]	2.33×10^{-17}
cg16327326	16:1662895	16p13.3	IFT140[TSS1500]	2.42×10^{-17}
cg20296188	17:77397067	17q25.3	HRNBP3[5′UTR]	2.47×10^{-17}
cg01235203	3:187461998	3q27.3	BCL6[5′UTR]	2.80×10^{-17}
cg01245787	4:71859630	4q13.3	DCK[1st Exon]	3.57×10^{-17}
cg00973947	3:143692255	3q24	C3 or f58[5′UTR, Body, 1st Exon]	7.95×10^{-17}
cg03406394	15:60296138	15q22.2	FOXB1[TSS1500]	1.26×10^{-16}
cg07279963	3:61546526	3p14.2	PTPRG[TSS1500]	1.49×10^{-16}
cg08133824	7:99698538	7q22.1	MCM7[Body, TSS200]; AP4M1[TSS1500]	1.66×10^{-16}
cg06656994	1:179713176	1q25.2	FAM163A[5′UTR]	2.60×10^{-16}
cg05396178	6:27833129	6p22.1	HIST1H2AL[1st Exon, 5′UTR]	4.06×10^{-16}
cg01438737	20:42086396	20q13.11	SFRS6[TSS200]	4.92×10^{-16}
cg22704520	2:200820451	2q33.1	C2 or f60[Body, 1st Exon, 5′UTR]; C2 or f47[5′UTR]	5.00×10^{-16}

TSSxxx = CpG is located within xxx bp of the transcription start site.

the role played by miRNA in the etiology of psychiatric disorders and add support to the growing interest for the use of miRNAs as possible biomarkers for such diseases. In this perspective, it has to be noted that, among our more significant results in the multivariate analyses, another CpGs (cg14035771) near a miRNA (*miR137*) was associated with severity of childhood maltreatment (see Table 4). miR137 is mainly found in hippocampus and amygdala and has been strongly associated with schizophrenia, but also with autism spectrum disorders (Pinto *et al.* 2014; Ripke *et al.* 2011; Yin *et al.* 2014). As for *miR124*, *miR137* has also been associated to amygdala functionality and emotion processing (Mothersill *et al.* 2014). Again, our results support the idea that the search of miRNAs and their regulatory networks in psychiatric diseases may prove to be fruitful in understanding the complex interplay of genes involved in these diseases.

More generally, our results are in agreement with several studies showing that BPD, certainly as a result of early life adversities, is definitely associated with broad epigenetic signatures involving several genes interacting together and genes that may be highlighted by a methylome-wide association study (Dammann *et al.* 2011; Groleau *et al.* 2014; Perroud *et al.* 2011, 2013; Steiger *et al.* 2013; Teschler *et al.* 2013; Thaler *et al.* 2014).

In a previous genome-wide methylation analysis, Teschler *et al.* found several specific CpG sites that exhibited increased methylation in BPD patients compared with controls (Teschler *et al.* 2013). But none of the CpGS or genes they found to be associated with BPD were found in the top of hits in our study. Several reasons might explain these discrepant results. First, the sample size in Teschler *et al.* (2013) study was very small consecutively increasing the risk of type 1 errors. Our study was adequately powered and may thus not have replicated possible false positive findings. Secondarily, the design of our study was different. Where Teschler *et al.* (2013) used 'normal controls', we decided to use MDD most of them suicide attempters thus controlling for not only MDD but also for history of suicide attempt. Thus, our results may truly be associated with BPD whereas Teschler *et al.* (2013) findings might highlight genes that could be associated with BPD but also with suicide attempt and/or MDD. Finally, although we were able to validate one of our findings, namely cg04927004 near *miR124-3*, using pyrosequencing method, the results were not as strong as the one found using the Illumina chip and we unfortunately did not validate other results. Thus, our findings should be considered with this caveat in mind and further studies should try to replicate them. Partly for the same reasons, we did not find any association with CpGs near or within *NR3C1* although epigenetic changes were previously reported in BPD subjects in several studies (Dammann *et al.* 2011; Martin-Blanco *et al.* 2014; Perroud *et al.* 2011; Steiger *et al.* 2013). We assessed more than 400 000 CpGs and only the more significant ones were discussed and reported in this study, thus CpGs near or within *NR3C1* might still show an association with BPD but at a lesser level than the one found to be significant.

Our study has several limitations including the self-measurement of childhood trauma and the absence of 'true' controls. Nevertheless our results make sense and our main finding (*miR124*) is also found in the case–control design of our study for which the assessment of subjects has been made by trained clinicians. Moreover the inclusion of MDD subjects should be observed as an advantage compared with controls coming from the general population, as they allow us to control for depression which is frequently found in BPD. Another main limitation is that we are not able to determine whether methylation signatures

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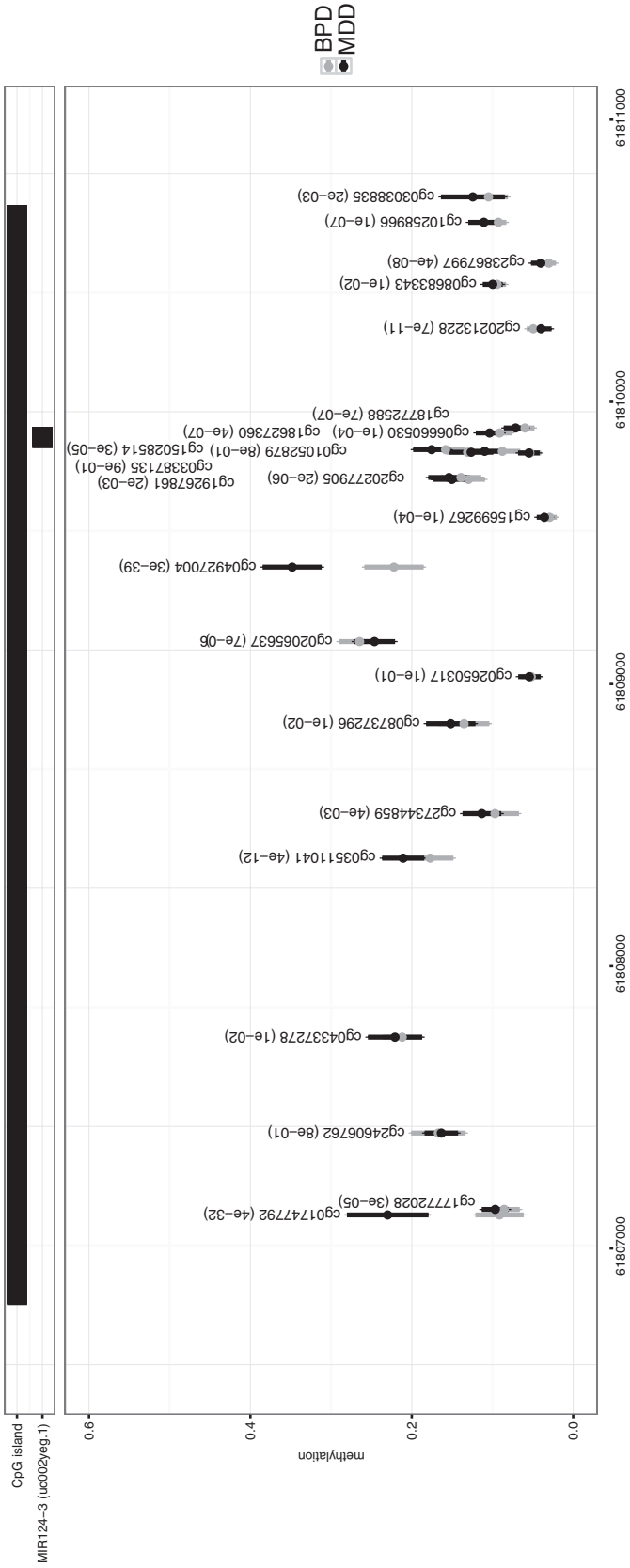

Figure 2: Overview of methylation pattern around hsa-miR-124-3 showing levels of methylation for each CpG for BPD and MDD subjects.

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Table 6: Result of querying the blood-sorted cell database for cg04927004

Gene		cg04927004 miR 124-3
Mean methylation in	CD8T	0.313920
	CD4T	0.220000
	NK	0.279240
	B cell	0.279360
	Mono	0.207440
	Gran	0.198400
Low		0.175730
High		0.446620
Range		0.270890
F stat		6.885100
P value		0.000220

The table gives the mean methylation in each type of cell, the minimum and maximum methylation values observed, and *f*-statistics of an analysis of variance (ANOVA) testing association of the site with blood composition. Bcell: B-cell; CD8T: CD8+ T-cell; CD4T: CD4+ T-cell; Gran: granulocyte; Mono: monocyte; NK: natural killer cell.

were associated to transcriptional changes. Nevertheless in a recent study, using Infinium HumanMethylation27 BeadChip, Teschler *et al.* (2013) found that differences in methylation levels correlated with changes in gene transcription, suggesting that methylation differences we detected in this genome-wide study may have functional consequences. Several other confounding factors may explain the observed differences in methylation status between BPD and MDD subjects such as history of psychopharmacological treatments, any psychoactive substance such as alcohol and other toxics, and any other major life events. BPD and MDD subjects did not differ in term of current medication (Table 1), we thus do not believe that this may have had a major impact on our findings. Concerning alcohol and substance use disorders, these two variables were significantly more often found in BPD subjects than MDD subjects and may thus have driven a part of the observed associations. Correcting for these variables, which are very often found in BPD patients, as part of their psychopathology, will be tautological and may have artificially led to lose some of the most interesting findings. We thus have not controlled for that and this has to be kept in mind when interpreting our results. In addition, severity of BPD was unfortunately not assessed and further studies should probably look at whether or not our findings correlate with the severity of the disorder. Another potential confounding factor is the contribution from the many different blood cell types. If by using the methods developed by Jaffe and Irizarry (Jaffe & Irizarry 2014), we were able to correct for different blood cell types, we unfortunately did not have the technical possibility to check whether the results obtained by the method truly reflect the distribution of blood cell types in our sample. Nevertheless this approach has been shown to be reliable and there is no reason to think that this should not be the case in our sample. Furthermore our goal was not to validate the approach of Jaffe and Irizarry, but to check for a potential bias of our experimental setup to blood cell type composition according to this approach. We think the question of the reliability of this method would have been more central if we had observed incoherencies. Finally, some of our findings are near the lower limit of detection of the Illumina Infinium HumanMethylation450. However, the use of a multivariate analysis selecting only those results for which strong beta values are obtained, thus explaining a large fraction of the observed difference removed CpGs for which too many individuals showed results near the lower limit of detection of the chip and thus leading to weak association with the outcome variable.

Table 7: List of known assay-validated targets for hsa-mir-124 according to mirTarBase

miRTarBase ID	miRNA	Gene	EG ID	Ref. PMID
MIRT001825	hsa-miR-124	BACE1	23621	18434550
MIRT004284	hsa-miR-124	CCL2	6347	19404929
MIRT004283	hsa-miR-124	CDK2	1017	19404929
MIRT002642	hsa-miR-124	CDK6	1021	17308079
MIRT002025	hsa-miR-124	CEBPA	1050	18451139
MIRT004726	hsa-miR-124	E2F6	1876	19843643
MIRT000478	hsa-miR-124	EFNB1	1947	20308325
MIRT003523	hsa-miR-124	ELK3	2004	20144220
MIRT002697	hsa-miR-124	IQGAP1	8826	19843643
MIRT005083	hsa-miR-124	NFKBIZ	64332	19502795
MIRT004546	hsa-miR-124	NR3C1	2908	19131573
MIRT001168	hsa-miR-124	NR3C2	4306	19944075
MIRT003051	hsa-miR-124	RDH10	157506	17724173
MIRT002616	hsa-miR-124	RELA	5970	15685193
MIRT002665	hsa-miR-124	SLC16A1	6566	19427019
MIRT004725	hsa-miR-124	SMYD3	64754	19843643
MIRT004724	hsa-miR-124	VIM	7431	19843643
MIRT002653	hsa-miR-124-3p	AHR	196	22024478
MIRT006451	hsa-miR-124-3p	AR	367	22386953
MIRT006479	hsa-miR-124-3p	EZH2	2146	21672940
MIRT006562	hsa-miR-124-3p	HMGA1	3159	22249617
MIRT006541	hsa-miR-124-3p	IL6R	3570	22153071
MIRT002731	hsa-miR-124-3p	ITGB1	3688	21112327
MIRT006478	hsa-miR-124-3p	ROCK2	9475	21672940
MIRT002716	hsa-miR-124-3p	SNAI2	6591	22253443

In summary, this study highlights the usefulness of whole methylation scans in order to find new possible targets that may explain the link between child maltreatment and adulthood psychopathologies. More specifically, we underlined the involvement of miRNAs in neurodevelopmental disorders such as BPD, which supports further the idea to investigate more deeply disruption of miRNA pathways in neuropsychiatric diseases. Furthermore, we suggest that these miRNAs may mediate the role of aversive events in the development of BPD and thus deserve further research. Better understanding of the epigenetic processes involving these genes may increase the comprehension of the disease and help identify therapeutic processes, including primary prevention of the late impact of child maltreatment. Further studies should definitely look at DNA methylation changes in BPD subjects and look at how in BPD subjects these epigenetic changes are associated with severity of childhood maltreatment.

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Acknowledgments

This study was funded by the National Center of Competence in Research (NCCR) 'SYNAPSY – The Synaptic Bases of Mental Diseases', pilot project to N.P. This study was supported by AXA Research Fund. We would like to thank Gérald Bouillault, Jean-Jacques Kunckler, Sophie Blin, Charlotte Germond, Venus Kaby, Karen Dieben and Caroline Waeber for their help in recruiting and assessing patients. Authors have no personal affiliations or financial relationships with any commercial interest to disclose relative to this article.

Supporting Information

Additional supporting information may be found in the online version of this article at the publisher's web-site:

Figure S1: Level of methylation in MDD and BPD patients at cg04927004. Results from the pyrosequencing ($P = 0.08$).

Figure S2: Correlation (Pearson's r) between beta values obtained from the Illumina chip and methylation values obtained from the pyrosequencing for cg04927004.

Table S1: DNA sequence for cg04927004.

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